



Experimental and modeling investigation of the low-temperature oxidation of *n*-heptane

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ABSTRACT

The low-temperature oxidation of *n*-heptane, one of the reference species for the octane rating of gasoline, was investigated using a jet-stirred reactor and two methods of analysis: gas chromatography and synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) with direct sampling through a molecular jet. The second method allowed the identification of products, such as molecules with hydroperoxy functions, that are not stable enough to be detected using gas chromatography. Mole fractions of the reactants and reaction products were measured as a function of temperature (500–1100 K), at a residence time of 2 s, at a pressure of 800 Torr (1.06 bar) and under stoichiometric conditions. The fuel was diluted in an inert gas (fuel inlet mole fraction of 0.005). Attention was paid to the formation of reaction products involved in the low-temperature oxidation of *n*-heptane, such as olefins, cyclic ethers, aldehydes, ketones, species with two carbonyl groups (diones), and ketohydroperoxides. Diones and ketohydroperoxides are important intermediates in the low-temperature oxidation of *n*-alkanes, but their formation have rarely been reported. Significant amounts of organic acids (acetic and propanoic acids) were also observed at low temperatures. The comparison of experimental data and profiles computed using an automatically generated detailed kinetic model is satisfactory overall. A route for the formation of acetic and propanoic acids was proposed. Quantum calculations were performed to refine the consumption routes of ketohydroperoxides toward diones.

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1. Introduction

The oxidation of *n*-heptane has already been the subject of many experimental and modeling studies [1–32]. *n*-Heptane is one of the two primary reference fuels that are used for the octane rating for gasoline, with an octane number of 0 [33]. This species is also a possible surrogate for the *n*-alkanes present in diesel fuels [34].

The oxidation of *n*-heptane has been studied previously in engines [1–5] and in several types of laboratory reactors such as shock tubes [6–11], rapid compression machines [12,13], jet-stirred reactors [14–16], flow reactors [17], and flames [18–22]. Most of these studies were carried out under conditions of high-temperature oxidation (typically above 800 K), and relatively little attention was paid to the low-temperature oxidation of *n*-heptane, especially regarding the characterization of oxygenated reaction products

[12,13,15–17]. Low-temperature oxidation of hydrocarbons is of importance for the development of diesel and homogenous charge compression ignition (HCCI) engines, as this particular chemistry triggers autoignition phenomena [35]; it is also responsible for other combustion phenomena such as cool flames and negative temperature coefficient (NTC) zones.

The overall reaction path diagram for the low-temperature oxidation of an alkane is described in Fig. 1 [35]. At low temperature (from 500 K for *n*-heptane), the alkyl radicals, which are derived from the initial *n*-alkane, add to molecular oxygen and form peroxyalkyl radicals (ROO[•]). These radicals then isomerize to hydroperoxyalkyl radicals (•QOOH) by internal transfer of an H atom. •QOOH radicals add again to molecular oxygen (this is called the second addition) to form •OOQOOH radicals. The •OOQOOH radicals isomerize and decompose very easily into ketohydroperoxides (OQOOH) + •OH (the H-atom on the carbon atom bearing the hydroperoxy group can be abstracted very easily). Ketohydroperoxides, which have hydroperoxy functions with very low O–O bond energy, decompose easily to OH and alkoxy radicals, allowing low-temperature chain branching. The

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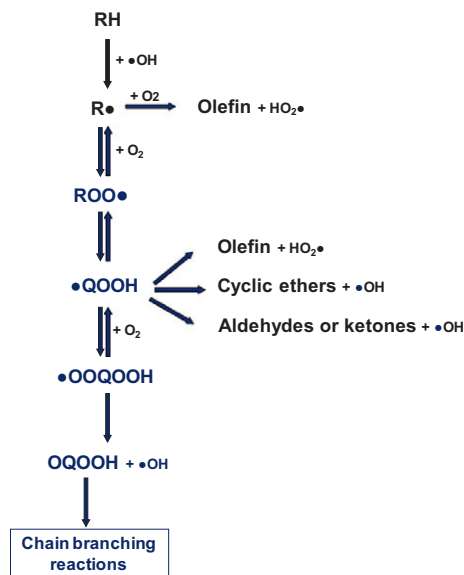


Fig. 1. Overall reaction path diagram of the low-temperature oxidation of an alkane (RH).

apparent activation energy of the sequence of reactions leading to chain branching is very low (close to 0 kcal mol⁻¹). At intermediate temperatures, •QOOH radicals can decompose to cyclic ethers, aldehydes, or ketones + •OH, or to olefins + HO₂•. These decomposition channels are chain-carrying and have an inhibiting effect on the reaction compared to the chain-branching channels. The apparent activation energy of the chain-carrying reactions is higher (about 20 kcal mol⁻¹). The inhibiting effect and the higher apparent activation energy of this intermediate-temperature mechanism are responsible for the NTC phenomenon that is observed for *n*-alkanes from about 650 K. At higher temperatures, alkyl radicals decompose by β-scission reactions leading to small species, which react with molecular oxygen (not shown in the scheme in Fig. 1). The branching is obtained from the decomposition of hydrogen peroxide (H₂O₂) and from the reaction of H-atoms and molecular oxygen, yielding •OH + •O. The apparent activation energy of the high-temperature mechanism is about 30 kcal mol⁻¹. This explains why the reactivity increases again in the temperature region following the NTC (from 800 K).

There are only a few experimental studies of the low-temperature oxidation of *n*-heptane with species analysis. Dagaut et al. [16] performed an experimental study of the oxidation of *n*-heptane in a jet-stirred reactor at temperatures ranging from 550 to 1150 K, at pressures from 1 to 40 atm, with residence times from 0.1 to 2 s and under stoichiometric conditions with highly diluted mixtures in nitrogen (inlet fuel mole fraction between 0.005 and 0.01). The fuel mole fraction profiles exhibited negative temperature coefficient (NTC) behavior, except at high pressure (40 atm), where it tended to disappear. The formation of many reaction products was observed during this study. Low-temperature species are mainly oxygenated species with the same skeleton as the reactant (1-heptene, 2-heptene, 3-heptene, 2-heptanone, 3-heptanone, heptanal, *cis*- and *trans*-2-methyl-5-ethyltetrahydrofuran, *trans*-2-methyl-4-propyloxetane, 2-propyltetrahydrofuran, 2-methyltetrahydrofuran, 2-butyloxirane, *cis*-2-methyl-4-propyloxetane, 2,4-diethyloxetane, 2-ethyltetrahydropyran, 2-ethyl-3-propyloxetane, *cis*- and *trans*-2-methyl-3-butyloxirane) and small species such as carbon monoxide, methanol, formaldehyde, ethylene, and propene. At high temperatures, the formation of oxygenated species with the same skeleton as the reactant was not observed any more and the main reaction products were unsaturated hydrocarbons (mainly olefins) and carbon oxides.

Lenhert et al. studied the low-temperature oxidation of *n*-heptane in a pressurized flow reactor (at 8 atm) [17], where strong NTC behavior was observed. Reaction products were quantified by gas chromatography. These products were carbon oxides, cyclic ethers (*cis*- and *trans*-2-methyl-5-ethyltetrahydrofuran, 2-propyltetrahydrofuran, and 2-methyl-4-propyloxetane), olefins (ethylene, propene, 1-butene, 1-pentene, 1-hexene, 1-heptene, *cis*- and *trans*-2-heptene, and *trans*-3-heptene), and aldehydes (acetaldehyde, propanal, acrolein, butanal, and pentanal).

Minetti et al. performed an experimental study of the oxidation of *n*-heptane in a rapid compression machine [12]. They measured ignition delay times of stoichiometric *n*-heptane–air mixtures at temperatures ranging from 600 to 900 K and at high pressure (3–9 bar). The obtained ignition delay time profiles versus temperature exhibit NTC behavior. Reaction products were analyzed using gas chromatography coupled to mass spectrometry. They observed the formation of species with the same skeleton as the reactant: heptene isomers and some cyclic ethers (2-methyl-5-ethyltetrahydrofuran, 2-propyltetrahydrofuran, 2-methyl-4-propyloxetane, and 2,4-diethyloxetane). Other reaction products were carbon monoxide, 1-olefins from ethylene to 1-hexene, acetaldehyde, propanal, butanal, methane, ethane, and propane.

The first modeling studies of the oxidation of *n*-heptane were carried out in the late 1970s and in the 1980s by Coats and Williams [23] (high-temperature oxidation only), and by Westbrook et al. [24,25]. In 1989, Westbrook et al. [25] proposed the first detailed kinetic model accounting for the low- and high-temperature oxidation of *n*-heptane. Since then, several low- and high-temperature oxidation models have been proposed for this species [26–32]. All these models were constructed using the commonly accepted low-temperature branched-chain mechanism via the formation of a degenerate branching agent (hydroperoxide species) for the oxidation of hydrocarbons (Fig. 1) [35]. These models were validated against experimental data available in the literature and were widely used to run numerical engine simulations (e.g., [36–38]).

The development of reliable detailed kinetic models relies on a good understanding of the reaction mechanism that occurs at low-temperature. Recently, an experimental study of the low-temperature oxidation of a small alkane, *n*-butane, was performed in a jet-stirred reactor using two methods of analysis: gas chromatography (in Nancy, France) and synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) with direct sampling through a molecular jet (in Hefei, China) [39]. Attention was paid to the identification and the quantification of reaction products such as organic acids, ketohydroperoxides (OQOOH species in Fig. 1, which are chain-branching agents responsible for the low-temperature oxidation of hydrocarbons), and products of decomposition of the ketohydroperoxides (species with two oxygenated functions, such as diones). The purpose of this paper is to extend the same approach to a larger *n*-alkane: *n*-heptane. The experimental results obtained with both methods of analysis have been confronted and experimental data have been compared with simulation results computed using a detailed kinetic model generated with software EXGAS [31].

2. Experimental methods

The oxidation of *n*-heptane was performed in a jet-stirred reactor, which can be modeled as a perfectly stirred reactor. The jet-stirred reactor used in this study was designed following the rules established by the team of Villermaux in the seventies [40,41]. The residence time distribution study performed by Matras and Villermaux [40] showed that good macro-mixing was obtained. Later annular preheating was added to the reactor to heat the reacting mixture in a very short period of time before it entered

the reactor. This ensures the homogeneity of the temperature in the gas phase [42]. A jet-stirred reactor working at pressures up to 10 bar was also developed by Dagaut et al. [43] by following the rules of Villermaux [40,41]. Dagaut et al. also verified that the macro-mixing was good and that the temperature of the gas phase was homogeneous, using a thermocouple [43]. Since then, this reactor has often been used for numerous gas phase kinetic studies of oxidation of hydrocarbons and oxygenated compounds [16,44–46].

Experiments were performed at a constant pressure of 1.06 bar, at a residence time of 2 s, at temperatures ranging from 500 to 1100 K in Nancy, and from 500 to 720 K in Hefei, under stoichiometric conditions. The fuel was diluted in helium in Nancy and in argon in Hefei (*n*-heptane mole fraction 0.005). Argon was used in Hefei because its signal (at a photoionization energy of 16.6 eV) is needed for the normalization of other signals.

The reactor, made of fused silica, consists of a quartz sphere (volume 95 cm³) into which diluted reactant enters through an injection cross located at its center. The gas jets provoke high turbulence, leading to homogeneity of the gas phase in composition and temperature. The reactor is operated at constant temperature and pressure and it is preceded by an annular preheating zone in which the gas temperature is increased up to the reactor temperature before entering. The residence time of the mixture inside the annular preheater is very short compared to its residence time inside the reactor (by a few percent). Both the spherical reactor and the annular preheating zone are heated by heating resistances rolled up around their walls in Nancy. In Hefei, because of the lateral cone, it was not possible to use heating resistances for the heating of the spherical part. A special oven with the shape of the reactor (sphere and cone) was designed and used for the heating. The reaction temperature was measured with a type K thermocouple inserted in the intra-annular part of the preheater (the extremity of the thermocouple is located at the center of the reactor in a glass finger in the injection cross). The accuracy of the reaction temperature with the two types of heating device was ± 2 K. Gas flow rates were controlled by mass flow controllers; Coriolis flow controllers were used for *n*-heptane. The uncertainty in the flow measurements is around 0.5% for each controller. This results in a maximum uncertainty of about 2% in the residence time.

In France, oxygen and helium were provided by Messer and their purities are reported as 99.995% and 99.999%, respectively. *n*-Heptane was provided by Sigma Aldrich (purity of 99%). In China, gases were provided by Nanjing Special Gas Factory Co., Ltd. Purities of argon and oxygen were 99.99% and 99.999%, respectively. *n*-Heptane was purchased from Aladdin Reagent Co., Ltd., with a purity of 99%.

Two methods were used for the analysis of the reaction products. Online gas chromatography was used in Nancy, whereas SVUV photoionization mass spectrometry with direct sampling through a molecular jet was used in Hefei. The first method provides the advantages of easy separation of different isomers, high sensitivity, and easy and direct quantification of species, but it is not suitable for analyzing unstable species such as hydroperoxides, as they decompose before the analysis. The main advantage of the second method is that there is a very short delay between the sampling in the reactor and the analysis using a time-of-flight mass spectrometer. The separation of the different isomers is possible but requires much more treatment. The quantification of a species requires knowledge of its photoionization cross section, which is not available for all species (these data can only be obtained by measuring the response of known amounts of standards when they are available).

2.1. Analytic method used in Nancy: gas chromatography

The outlet products were analyzed online by gas chromatography. The online analysis of products that are liquid under standard

conditions was made possible by a heated transfer line between the reactor outlet and the chromatograph sampling gate, which was also heated. The temperature of the transfer line can be set between ambient temperature and 473.15 K maximum. During the study, the temperature of the transfer line was set to 373.15 K. This temperature was high enough to avoid condensation of species. Three gas chromatographs were used for the quantification of the different species.

The first gas chromatograph, equipped with a Carbosphere packed column, a thermal conductivity detector (TCD), and a flame ionization detector (FID), was used for the quantification of O₂, CO, CO₂, methane, ethylene, acetylene, and ethane. The temperature profile was 303.15 K for 10 min—ramp of 5 K min^{−1} up to 573.15 K—573.15 K for 16 min. The carrier gas was helium (25 mL min^{−1}, constant pressure, splitless). The second gas chromatograph was fitted with a PlotQ capillary column and a FID, and was used for the quantification of molecules from methane to reaction products containing up to 5 carbon atoms and 1 or 2 oxygen atoms. The temperature profile was 333.15 K for 10 min—ramp of 5 K min^{−1} up to 623.15 K—623.15 K for 15 min. The carrier gas was helium (2 mL min^{−1}, constant flow, split 1/10). The third gas chromatograph was fitted with a HP-5 capillary column and a FID and was used for the quantification of molecules that contain at least five carbon atoms. The temperature profile was 313.15 K for 30 min—ramp of 5 K min^{−1} up to 573.15 K—573.15 K for 8 min. The carrier gas was helium (1 mL min^{−1}, constant flow, split 1/10).

Calibration was performed by injecting standards when available or by using the effective carbon number method (only applicable with the flame ionization detector) when standards were not available. Maximum relative uncertainty in mole fractions was estimated as $\pm 10\%$ and the limit of detection for species was about 1 ppm for species analyzed using the FID and 100 ppm for species analyzed using the TCD.

The identification of reaction products was performed using a gas chromatograph equipped with a PlotQ or an HP5 capillary column and coupled to a mass spectrometer. The mass spectra of all detected reaction products were included in the NIST 08 Mass Spectra Database.

2.2. Analytic method used in Hefei: synchrotron vacuum ultraviolet photoionization mass spectrometry with direct sampling through a molecular jet

Species from the reactor were analyzed by a reflectron time-of-flight mass spectrometer (RTOF MS) with photoionization by synchrotron radiation. The reactor was coupled to the low-pressure photoionization chamber through a lateral fused silica conelike nozzle, which was inserted into the spherical part. The tip of the cone was pierced by a 50- μ m orifice. A nickel skimmer with a 1.25-mm-diameter aperture was located 15 mm downstream from the sampling nozzle. The sampled gases formed a molecular beam, which was intersected perpendicularly with synchrotron vacuum ultraviolet light. The ion signal was then detected with the RTOF MS, which was installed in the photoionization chamber vertically. Additional details about this experimental apparatus are available in [39].

The quantification of species is more difficult than in gas chromatography; it requires knowledge of the cross sections. This is only possible for species that are available for purchase. The relative uncertainty in the mole fractions is $\pm 25\%$ for these species.

3. Experimental results

Figures 2–9 display the evolution of the mole fractions of the reactants and of the reaction products with temperature. Some

species quantified by both methods of analysis show good agreement between the two sets of experimental data (*n*-heptane and oxygen in Fig. 2, ethylene and propene in Fig. 3, acetaldehyde in Fig. 4). Experimental data obtained in Hefei allowed the measurement of the mole fraction profiles of species that were not quantified in Nancy (water in Fig. 3, methanol and formaldehyde in Fig. 4, ketohydroperoxides in Fig. 10).

3.1. Analyses using gas chromatography (Nancy)

Numerous species were analyzed by online gas chromatography: *n*-heptane, oxygen, and 45 reaction products. Reaction products can be classified as follows:

- *Light species*: carbon oxides, methane, ethane, acetylene, ethylene (Fig. 3).
- 1-Olefins from propene to 1-heptene (Figs. 3, 6 and 7).
- Other unsaturated linear hydrocarbons: 2-heptene, 1,3-butadiene, 1,3-pentadiene, 1,4-heptadiene (Figs. 6 and 7).
- *Saturated and unsaturated aldehydes and ketones*: acetaldehyde, acrolein, propanal, acetone, methyl vinyl ketone, butanal, butanone, 2-butanone, 1-penten-3-one, 2-pentanone, pentanal, 2-hexanone, 1-hexen-3-one, 2-heptanone, 3-heptanone (Figs. 4, 5 and 9).
- *C₇ cyclic ethers*: 2-propyltetrahydrofuran, 2-ethyl-5-methyltetrahydrofuran, 2-methyl-4-propyloxetane, 2,4-diethyloxetane, 2-ethyl-3-propyloxirane, 2-butyl-3-methyloxirane, and some unsaturated cyclic ethers (Fig. 8).

Note that the formation of heptanal was not observed in this study, whereas it was in the study by Dagaut et al. [16]. They also observed the formation of other cyclic ethers such as 2-methyltetrahydropyran, 2-butyloxirane, and 2-ethyltetrahydrofuran. These were not observed in this study, probably because their mole fractions were too low (under the detection limit).

Some species, which have never been observed in previous oxidation studies as reaction products, have been detected in the present study. These species are organic acids—acetic and propanoic acids (Fig. 5), C₇ species with two carbonyl functions (or diones)—2,4-heptadione and 3,5-heptadione (Fig. 10), dihydro-5-methyl-2(3H)-furanone, and a species that could be tetrahy-

dro-5-methyl-2-furanmethanol (Fig. 8). Note that Kurman et al. [47] also observed the formation of dihydro-5-methyl-2(3H)-furanone in the low-temperature oxidation of *n*-dodecane. The formation of 3-heptene was expected [16] but this species was not observed; the corresponding peak is likely to be masked by another larger one (e.g., the peak of *n*-heptane).

Two types of mole fraction profiles were observed for these species at low temperature (500–800 K). For some of them (e.g., carbon monoxide, 1-olefins, cyclic ethers), the mole fraction increases and decreases smoothly with the temperature with a maximum at 650 K. For some others (e.g., organic acids, aldehydes and ketones such as propanal and acetone, and diones), the mole fraction increases abruptly with the start of reactivity around 525 K, with a sharp peak at 550 K. The first type of species are known to be stable products formed in the propagation steps at low and intermediate temperatures, and their mole fractions are somewhat proportional to the conversion of the reactant. Organic acids, aldehydes, ketones, and diones are also stable species and the sharp peak profiles were not expected at all. The cause for this unexpected sharp peak is discussed later.

Carbon atom balances were calculated at each studied temperature (the numbers are given in the Supplemental data). The relative difference in the expected and experimental numbers of carbon atoms at the outlet of the reactor is $\pm 10\%$ except in the low-temperature region, where it is $\pm 20\%$. The larger difference in the low-temperature region is due to the nonquantification of some oxygenated species.

3.2. Analyses using SVUV photoionization mass spectrometry (Hefei)

Experiments performed in Nancy were repeated in Hefei, but using SVUV photoionization mass spectrometry for the analysis. Figure 11 displays two mass spectra obtained at 530 and 650 K (photon energy 9.5 eV). Mass spectra were recorded at several photon energies: 9.5, 10, 10.5, 11, 13, and 16.6 eV. Masses were scanned from 28 to 146 amu for photon energies of 9.5, 10, 10.5, and 11 eV, and from 15 to 146 amu for photon energies of 13 and 16.6 eV. Figures 12 and 13 display the photoionization efficiency spectra of some selected masses between 56 and 146 amu. The changes in the slope in these spectra allow the identification of species by their ionization energy (IE). Table 1 gives the photoionization energies for the species that were detected using gas chromatography. The values given in this table come from the NIST database [48].

At 530 K, the main peaks were observed for m/z 72, 98, 100, 112, 114, 128, and 146 (Fig. 11). Note that species with m/z 72, 98, 100, 114, and 128 have been observed in the gas chromatography analyses (Table 1), whereas no species with m/z 112 or 146 was detected. The peaks at m/z 72, 98, 100, 112, 114, and 128 are still observed in the mass spectrum obtained at 650 K (Fig. 11). The peak at m/z 146 is not present any more, meaning that ketohydroperoxides are not detected at this higher temperature. There are new large peaks in the mass spectrum obtained at 560 K. These peaks are at m/z 56, 70, 84, 85, 86, 99, and 126. Note that species with m/z 56, 70, 84, and 86 were also detected in the gas chromatography analyses (Table 1), but that there were no species with m/z 85, 99, or 126. Peaks at m/z 70, 84, and 126 are likely to correspond to unsaturated species that are derived from the species with m/z 72 (butanone and butanal), 86 (C₅H₁₀O isomers: pentanal and pentanones), and 128 (C₇H₁₂O₂ diones). Peaks at m/z 85, 99, and 126 may correspond to fragments from larger ions.

Species assignment was performed using the photoionization efficiency spectra of the detected m/z (Figs. 12 and 13). Ionization energies of the possible species are indicated in the photoionization efficiency spectra in Figs. 12 and 13. Below we discuss the species that these peaks likely correspond to.

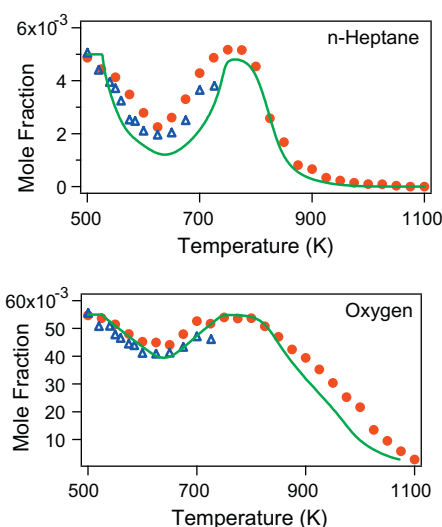


Fig. 2. Mole fractions of the reactants. ●: Experimental results obtained in Nancy. △: Experimental data obtained in Hefei (the signals of *n*-heptane and oxygen obtained at 10.5 and 13 eV, respectively, were calibrated using the experimental data obtained in Nancy). —: Simulation.

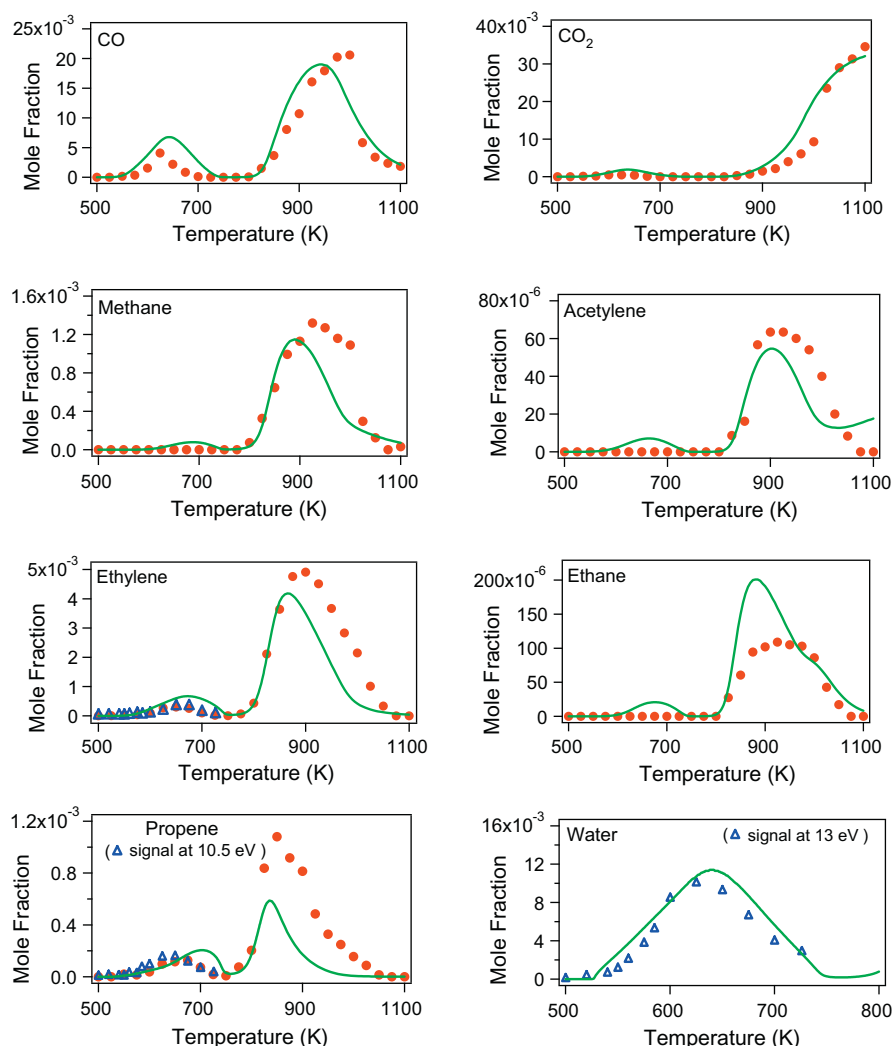


Fig. 3. Mole fractions of carbon oxides, C_1 – C_3 hydrocarbons, and water. ●: Experimental results obtained in Nancy. △: Experimental data obtained in Hefei (mole fractions of propene, ethylene and water were calculated from the signals obtained at 10.5, 11, and 13 eV, respectively). —: Simulation.

- m/z 56: We were not able to find any species that certainly corresponds to this peak, as the experimental IE (about 9.2 eV) is lower than the IEs of the two species observed using gas chromatography for this mass: 1-butene and acrolein (IEs of 9.55 and 10.11 eV, respectively [48]). Note that this peak can also stand for *iso*-butene and 2-butene, which have lower ionization energies (IE of 9.22 and 9.11 eV, respectively [48]), but which have not been observed in the gas chromatographic analyses.
- m/z 70: 2-butenal and 3-buten-2-one (IEs of 9.75 and 9.65 eV [48]), or it could also correspond to 1-pentene (IE of 9.49 eV [48]).
- m/z 72: Butanal and butanone (IEs of 9.82 and 9.52 eV, respectively).
- m/z 84: 2-pentenal, 1-penten-3-one, and 3-penten-2-one (IEs of 9.70, 9.50 and 9.39 eV, respectively [48]). It could also be 1-hexene (IE of 9.44 eV [48]).
- m/z 85: Probably a C_6H_{13} fragment coming from the decomposition of larger ions.
- m/z 86: Pentanal and 2-pentanone (IEs of 9.38 and 9.74 eV, respectively [48]). This peak could also correspond to 3-pentanone (IE of 9.31 eV [48]), which was not detected by gas chromatography.
- m/z 98: Heptene isomers (IEs of 9.35, 8.84, and 8.77 eV for 1-heptene, 2-heptene, and 3-heptene, respectively).
- m/z 99: Probably a C_7H_{15} fragment coming from the decomposition of larger ions. This peak cannot correspond to heptyl radical isomers, which would be derived from the reactant, as their IEs (e.g., 7.90 eV for the 1-heptyl radical [48]) are much lower than the experimental value (about 8.8 eV) in Fig. 12.
- m/z 100: $C_6H_{12}O$ isomers (IEs of 9.72, 9.35, and 9.12 eV for hexanal, 2-hexanone, and 3-hexanone, respectively [48]). Note that this peak does not take into account the signal for *n*-heptane at photon energies lower than 9.5 eV (*n*-heptane IE is 9.93 eV [48]).
- m/z 112: $C_7H_{12}O$ isomers (heptenones, heptenal, or cyclic ethers with one double bond) resulting from the decomposition of $C_7H_{14}O$ species, or it could also correspond to a fragment from a larger ion.
- m/z 114: $C_7H_{14}O$ isomers (species with the same skeleton as the reactant) such as heptanal, heptanone isomers, and cyclic ethers. The literature [48] gives the IEs for heptanal, 2-heptanone, 3-heptanone, and 4-heptanone (9.65, 9.3, 9.1, and 9.1 eV, respectively). Since there are no data in the literature for $C_7H_{14}O$ cyclic ethers, zero-point-energy-corrected adiabatic ionization energies have been calculated from the composite CBS-QB3

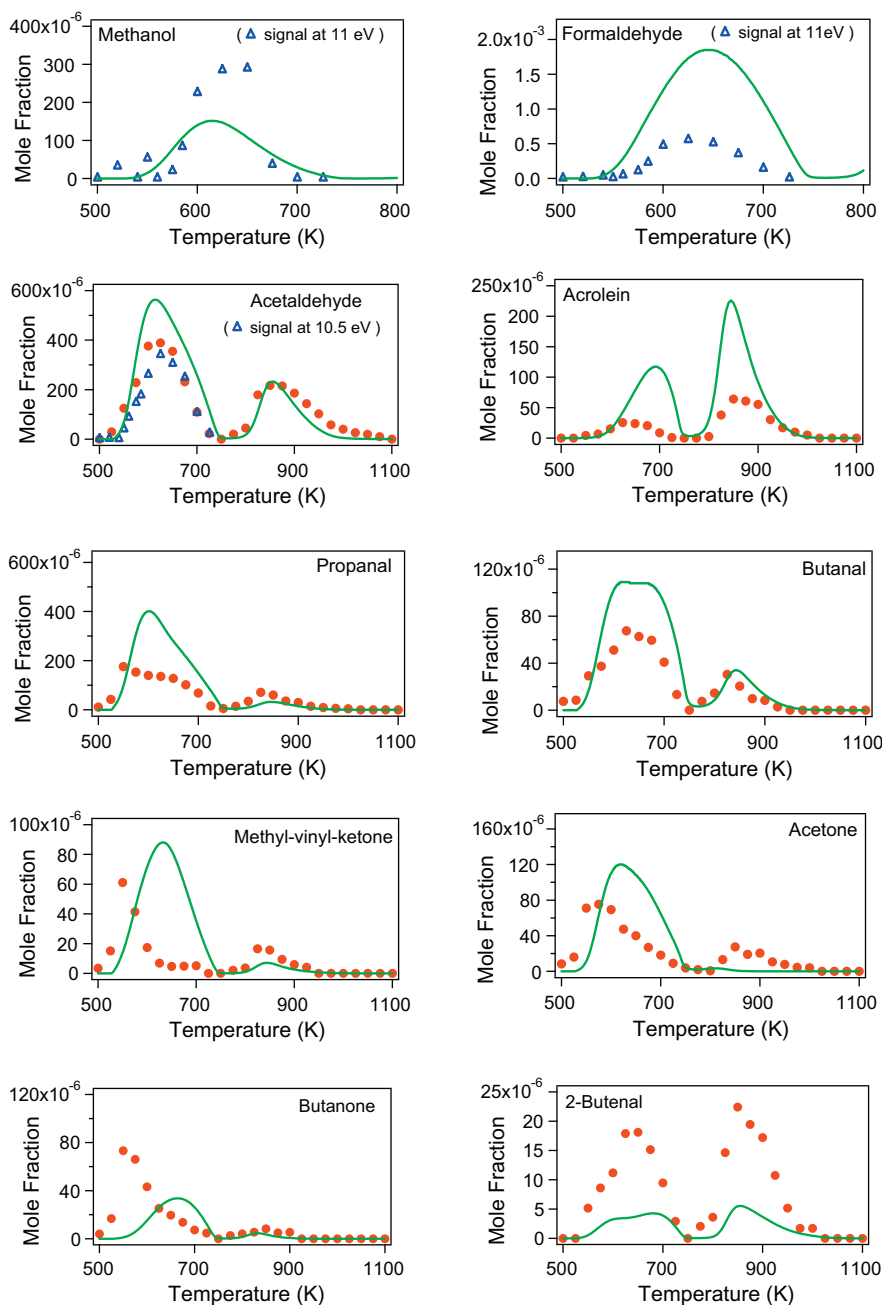


Fig. 4. Mole fractions of C_1 – C_6 oxygenated products (Part I). ●: Experimental results obtained in Nancy. △: Experimental data obtained in Hefei (mole fractions of methanol and formaldehyde were calculated from the signals obtained at 11 eV and the one of acetaldehyde from the signal obtained at 10.5 eV). —: Simulation.

method [49] using Gaussian03 [50]. Calculated ionization energies are 9.12, 9.05, and 9.08 for *n*-propyltetrahydrofuran, *trans*-1-methyl-4-ethyltetrahydrofuran, and *cis*-1-methyl-4-ethyltetrahydrofuran, respectively (Table 2). Note that there is almost no difference in the ionization energies of the *cis* and the *trans* isomers of 1-methyl-4-ethyltetrahydrofuran. All these calculated and experimental ionization energies for $C_7H_{14}O$ isomers are in good agreement with the experimental value for m/z 114 (Fig. 13).

- m/z 126: No literature data available. These species are probably unsaturated molecules deriving from the decomposition of diones (m/z 128) and of ketohydroperoxide isomers (m/z 146).
- m/z 128: Diones. As for $C_7H_{14}O$ cyclic ethers, there is no data in the literature for the IEs of these species. Theoretical calculations were performed for the three most probable

$C_7H_{12}O_2$ isomers: 1,3-heptadione, 2,4-heptadione, and 3,5-heptadione. These species have the two carbonyl functions separated by a CH_2 group and are expected to be formed in larger amounts than other isomers because the isomerizations of $ROO\cdot$ radicals into $\cdot QOOH$ radicals through six-membered ring transition states are the easiest isomerization channels [31,32]. Computed ionization energies are 9.32, 9.40, and 9.37 eV for 1,3-heptadione, 2,4-heptadione, and 3,5-heptadione, respectively (Table 2). In the photoionization efficiency spectra of mass 128, a change in the slope is visible at an IE of 9.3–9.4 eV, which is in agreement with the values calculated for diones.

- m/z 146: Ketohydroperoxide isomers. As for diones, there are no data in the literature for these species, and calculations were performed for seven different isomers (for diones, the most

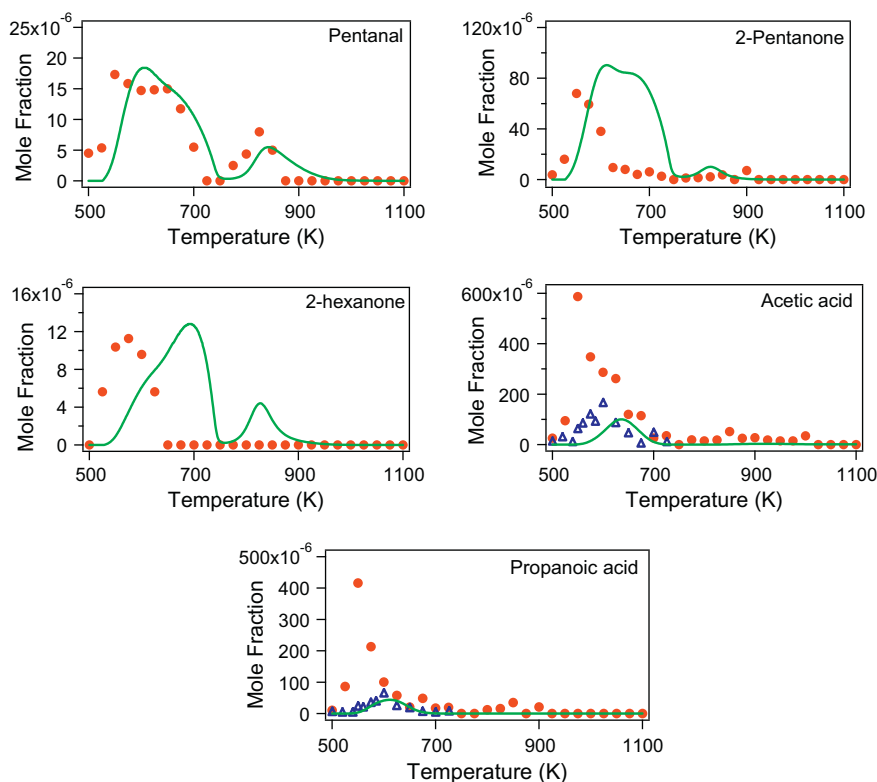


Fig. 5. Mole fractions of C_1 – C_6 oxygenated products (Part II). ●: Experimental results obtained in Nancy. △: Experimental data obtained in Hefei (mole fractions of organic acids were calculated from the SVUV-PIMS signals obtained for m/z 60 and 74 at 11 eV). —: Simulation.

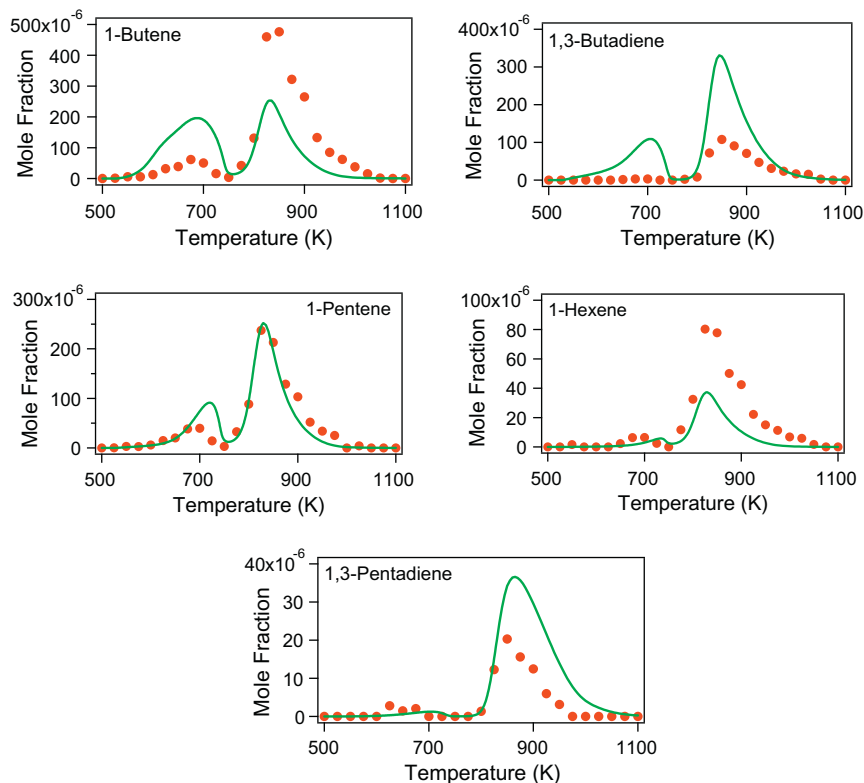


Fig. 6. Mole fractions of C_4 – C_6 unsaturated hydrocarbons. ●: Experimental results obtained in Nancy. —: Simulation.

probable isomers have the carbonyl and the hydroperoxide functions separated by a CH_2 group). Computed ionization energies are in the range 8.87–10.65 eV (Table 3). Note that

the ionization energies of the most probable isomers (3-heptanone-5-hydroperoxide, 1-heptanal-3-hydroperoxide, and 2-heptanone-4-hydroperoxide) are 9.31, 9.32, and

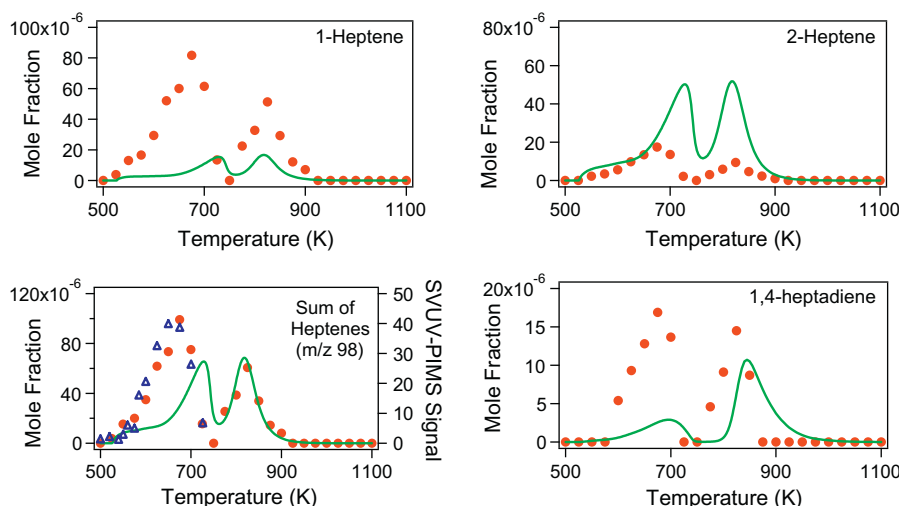


Fig. 7. Mole fractions of C₇ unsaturated hydrocarbons. ●: Experimental results obtained in Nancy. △: Experimental data obtained in Hefei (the signal for *m/z* 98 was obtained at 10 eV). —: Simulation.

9.57 eV, respectively. These computed ionization energies are in agreement with the experimental photoionization efficiency spectra of mass 146 (Fig. 13).

Two other important peaks were observed at photon energies higher than 11 eV for *m/z* 60 and 74. These correspond to acetic acid and propanoic acid, respectively (IEs are 10.65 and 10.44, respectively [48]). The scan in photon energy was carried out up to 10.4 eV, which is lower than the ionization of these two species, and it was not possible to have a clear identification according to the ionization energies. Note that these two species were also observed in the gas chromatographic analyses (Fig. 5).

3.3. Quantification of species using SVUV photoionization mass spectrometry

The experimental data obtained in Hefei were used to quantify species that were not detected using gas chromatography. Some species, which were detected by both methods (e.g., ethylene in Fig. 3), have been quantified using both sets of experimental results to verify the consistency of the two sources of data. The two reactants (*n*-heptane and oxygen) were directly calibrated by measuring the signal as a function of the mole fraction in the reactor in unreactive conditions.

The quantification of reaction products was performed and using the reactant signal, which was calibrated, as a reference [51], with the equation

$$x_i(T) = x_{ref}(T) \times \frac{S_i(T)}{S_{ref}(T)} \times \frac{\sigma_{ref}(E)}{\sigma_i(E)} \times \frac{D_{ref}}{D_i}, \quad (1)$$

$x_i(T)$ and $S_i(T)$ are the mole fraction and the signal of species *i* at temperature *T*, respectively. $\sigma_i(E)$ is the photoionization cross section of species *i* at the photon energy *E*. D_i is the mass discrimination factor of species *i*.

Ethylene was quantified using the *n*-heptane signal at 11 eV as a reference. Cross sections used for the quantification are from Cool et al. [52] and from Zhou et al. [53] for ethylene and *n*-heptane, respectively. Mole fractions calculated from the data obtained in Hefei are in very good agreement with the mole fractions obtained using gas chromatography (Fig. 3). Acetaldehyde and propene were quantified using the same method, but using the *n*-heptane signal at 10.5 eV as a reference. Cross sections used for the quantification are from Person and Nicole [54] for propene and from Cool et al. [55] for acetaldehyde. Again, mole fractions obtained using both

methods are in very good agreement (Figs. 3 and 4). Methanol and formaldehyde could not be quantified in Nancy. Their mole fractions were calculated from the *n*-heptane signal at 11 eV (Fig. 4). Cross sections used for the quantification are from Person and Nicole [56] for methanol and from Cooper et al. [57] for formaldehyde. Signals of *n*-heptane and of oxygen could not be used as references for the quantification of water (ionization energy of 12.621 eV [48]) because their cross sections are not known with accuracy in this photon energy region (the cross section of oxygen varies a lot—it is 8.63 Mb at 12.96 eV, 41.36 Mb at 12.97 eV, and 1.69 Mb at 13.05 eV [58])—and the cross section of *n*-heptane was not measured above 11 eV). The quantification was then performed using the signal of argon at the photon energy of 16.6 eV. The cross section of argon was from Samson and Stolte [59]. The mole fraction profile of water is displayed in Fig. 3.

3.4. Formation of diones and ketohydroperoxides

It was not possible to quantify diones (*m/z* 128) and ketohydroperoxides (*m/z* 146) directly from the SVUV photoionization mass spectrometry signals, as the cross sections of these species are not known (these species are not available for purchase). Therefore we compared the shapes of the signals obtained using SVUV photoionization mass spectrometry and the shapes of the mole fraction profiles obtained using gas chromatography for diones. The comparison in Fig. 10 shows that the profiles obtained using the two analytical methods are different. The signal obtained for diones using SVUV photoionization mass spectrometry has the shape that is usually observed for stable species, with no abrupt peak when the reaction starts, whereas the mole fraction profile obtained using gas chromatography exhibits an unexpected sharp peak. Note that the peak that is visible in the mole fraction profile of diones is very similar to the profile obtained for *m/z* 146 (ketohydroperoxides) in Fig. 10, which is the typical shape observed for branching agents, as was the case during the study of the oxidation of *n*-butane using SVUV photoionization mass spectrometry [39]. This led us to think that the sharp peak is likely due to the decomposition of ketohydroperoxides in the heated transfer line between the reactor and the gas chromatographs or in the heated injector of gas chromatographs just before the analysis. At higher temperatures, ketohydroperoxides are already decomposed before they exit the reactor and the diones that are detected are only formed due to the decomposition of ketohydroperoxides in the reactor.

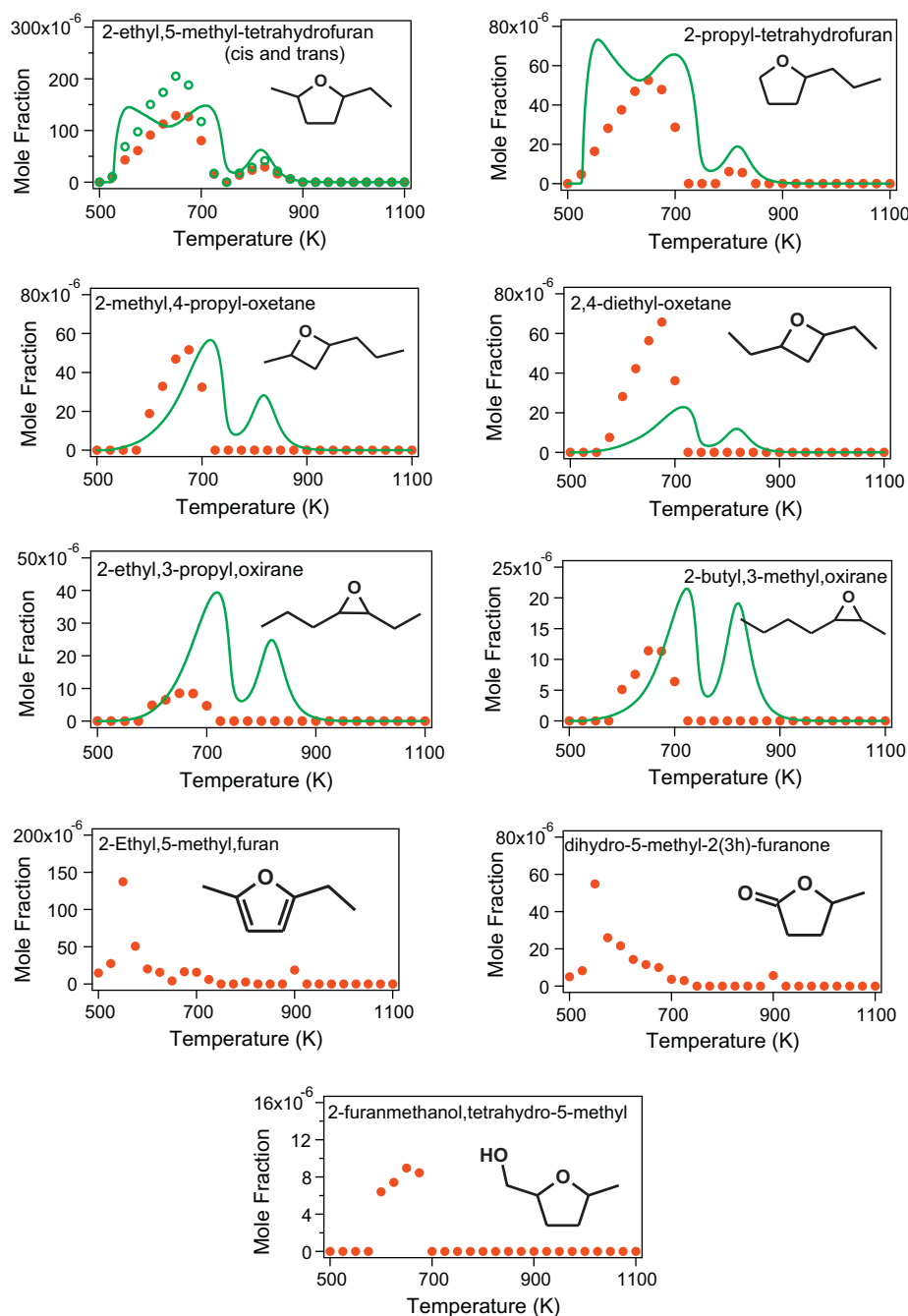


Fig. 8. Mole fractions of C_7 cyclic ethers. ● and ○: Experimental results obtained in Nancy. —: Simulation.

Thanks to these experimental observations, we can conclude that

- The decomposition of ketohydroperoxides leads to diones and some other oxygenated species.
- The main isomers for ketohydroperoxides are those with the two groups in positions 2, 4 and 3, 5, as we obtained mainly 2,4- and 3,5-heptadiones. This is in agreement with the low-temperature oxidation chemistry of *n*-alkanes.
- The mole fraction of ketohydroperoxides is greater than 1.3×10^{-4} in the peak region (the value is deduced from the sum of the mole fractions of 2,4- and 3,5-heptadiones).

3.5. Formation of organic acids

The formation of organic acids has rarely been observed in combustion experiments. Acetic acid has already been detected in the study of the oxidation of *n*-butane in a jet-stirred reactor [39] and in the study of the oxidation of *n*-butylcyclohexane in a flow reactor [60].

As they were not available in the literature, cross sections of acetic and propanoic acids were measured in the present work at photon energies ranging from 8.7 to 11.7 eV using pure standards. The evolution of the cross sections of these species as a function of the photon energy is displayed in Fig. 14. Note that experimental

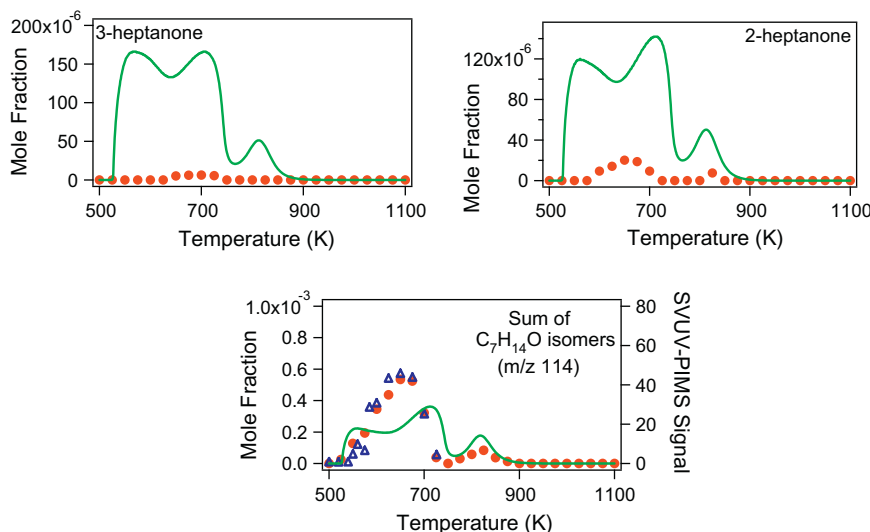


Fig. 9. Mole fractions of C_7 species with one oxygen atom. ●: Experimental results obtained in Nancy. △: Experimental data obtained in Hefei (the signal for m/z 114 was obtained at 10 eV). —: Simulation.

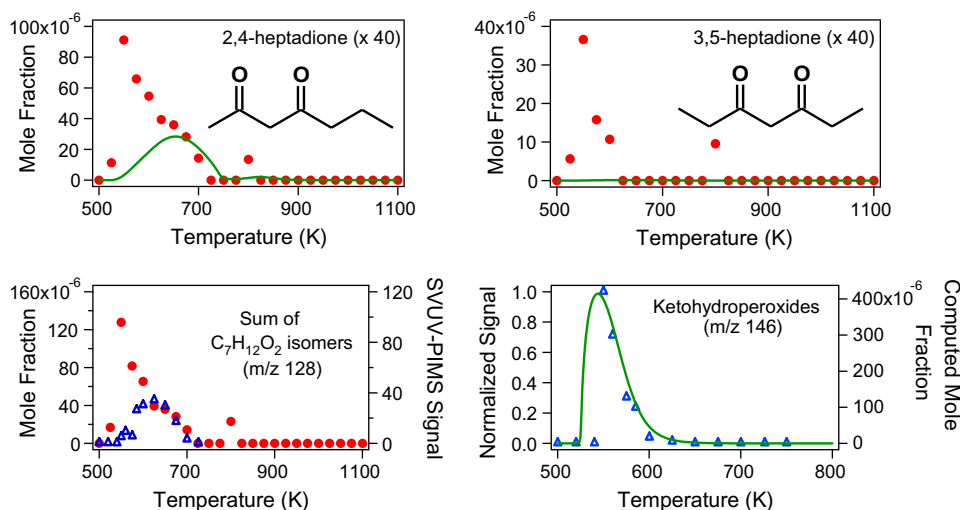


Fig. 10. Mole fractions of C_7 species with two or three oxygen atoms. ●: Experimental results obtained in Nancy. △: Experimental data obtained in Hefei (the signal for m/z 128 and m/z 146 were obtained at 10 and 9.5 eV, respectively). —: Simulation.

ionization energies are in very good agreement with data from the literature (10.44 and 10.65 eV for propanoic acid and acetic acid, respectively [48]).

Mole fractions of acetic and propanoic acids were calculated from the n -heptane signal at 11 eV using the cross sections measured in this work (Fig. 5). As for diones, it can be seen in Fig. 5 that data obtained with the two types of analyses are different: GC data exhibit a sharp peak located at 550 K, whereas there is no sharp peak in the SVUV-PIMS data, for which the maximum mole fractions are obtained around 600 K. Again, in the Nancy experiments, the higher concentration of these species results from the decomposition of ketohydroperoxides in the transfer line between the reactor and the gas chromatographs. On the other hand, the relatively lower formation of these acids in the Hefei experiments can certainly be trusted, as the probability of reactions in the molecular beam is very limited.

4. Discussion

The experimental results obtained in this study were compared with data computed using a detailed kinetic model generated using the software EXGAS [30].

4.1. Model description

This model contains all types of elementary reactions that are involved in the low- and high-temperature oxidation chemistry of n -alkanes. It is made of three parts. The first part is a C_0 – C_2 reaction base that contains the reactions of the small species having two carbon atoms or fewer. The second part is a comprehensive primary mechanism containing the reactions of the reactant and of the radicals deriving from the reactant having more than two carbon atoms (the reactions of radicals having two carbon atoms or fewer are already in the C_0 – C_2 reaction base). The molecular products formed in the primary mechanism are lumped to reduce the size of the model (lumping consists in gathering the molecular products having the same global formulae and the same functional groups into one generic species). The third part is a global secondary mechanism that contains the reactions of the lumped primary products. The reactions that consume these products are written in a global way in order to promote the formation of species that are already included in the primary mechanism and in the C_0 – C_2 reaction base. As an example, the lumped $C_7H_{14}O_3$ ketohydroperoxide will decompose in one step into a CO molecule, an OH radical, an n -propyl radical, and a propanal molecule [31]. The rate constant

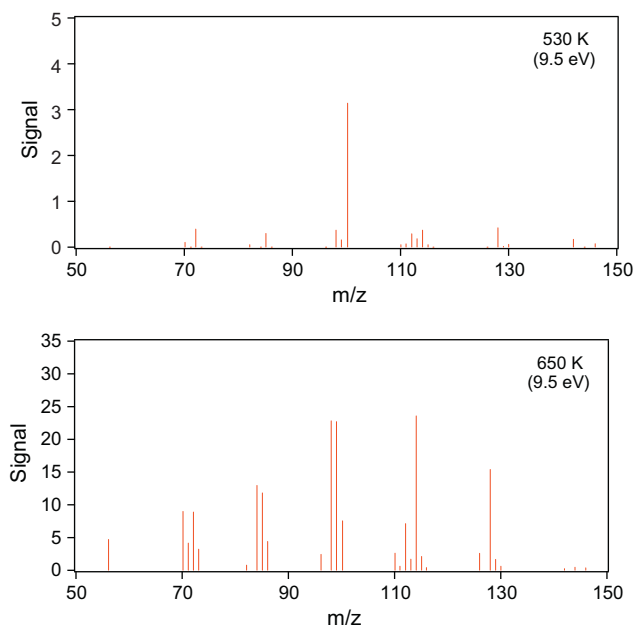


Fig. 11. Mass spectra obtained during experiments performed at 530 K and 650 K with a photon energy of 9.5 eV.

for this global reaction is the one corresponding to the breaking of the weak O–O bond.

The initial model generated with EXGAS contained 303 species involved in 1994 reactions. Given the numerous reaction products detected during this experimental part of this work, modifications were made in the model to better account for experimental observations:

- Lumped species were considered individually when they were quantified in experiments. This is the case for heptane isomers, heptanones, and $C_7H_{14}O$ cyclic ethers.
- Reactions accounting for the formation of diones, acetic acid, propanoic acid, methyl vinyl ketone, 2-butenal, 1,3-pentadiene, 1,4-heptadiene, and 2-hexanone have been added.

All these new reactions are presented in the next sections.

4.2. Modeling of the consumption of ketohydroperoxides and of the formation of diones

In the initial model, all $C_7H_{14}O_3$ ketohydroperoxides were lumped into one generic species and the decomposition of this species was considered in one step forming small species. Some modifications in the formation and the consumption of this species were made to try to reproduce the formation of intermediates that could be derived from ketohydroperoxides such as diones. As 2,4-heptadione and 3,5-heptadione were the only diones observed in the experiments, only ketohydroperoxides leading to these diones were considered individually (the ketohydroperoxide species leading to 1,3-heptadione was also considered individually, although 1,3-heptadione was not observed in the experiments). In the initial

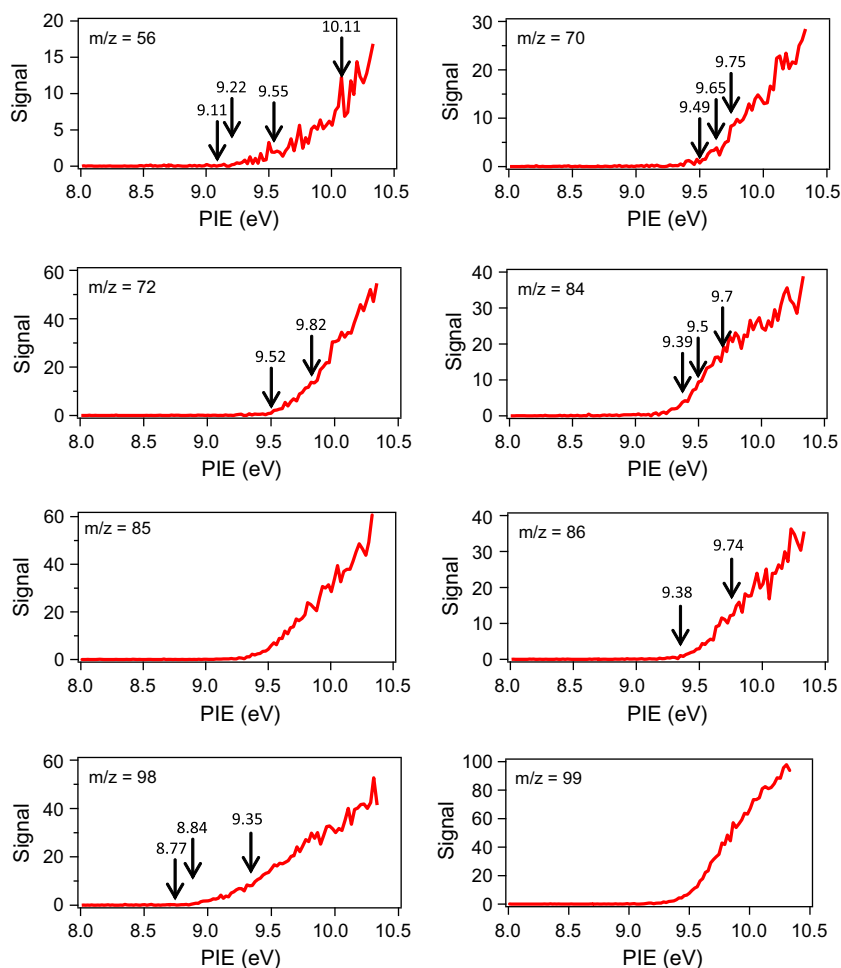


Fig. 12. Photoionization efficiency spectra of mass 56, 70, 72, 84, 85, 86, 98, and 99 sampled from the reactor. The temperature in the reactor was 600 K.

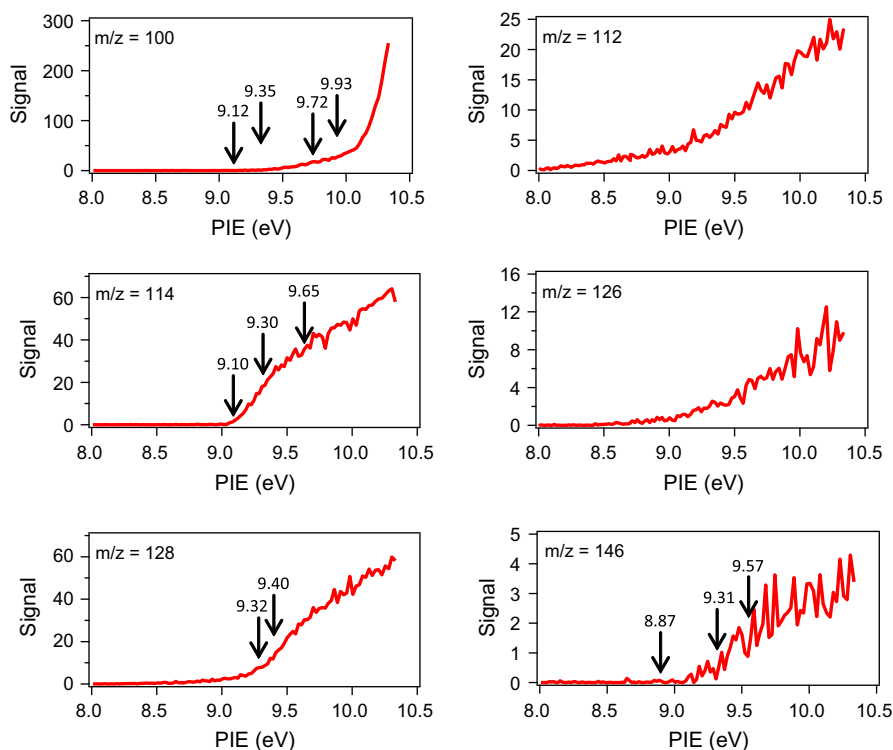


Fig. 13. Photoionization efficiency spectra of mass 100, 112, 114, 126, 128, and 146 sampled from the reactor. The temperature in the reactor was 600 K.

model, the lumped $C_7H_{14}O_3$ ketohydroperoxide decomposes into small species through a global reaction and the kinetic constant was that of the breaking of the weak O–O bond. Quantum calculations were performed in order to find other possible pathways for the consumption of ketohydroperoxides and of $C_7H_{13}O_2$ alkoxy radicals. To reduce CPU time, these calculations were performed for $C_5H_{10}O_2$ and $C_5H_9O_2$ species using Gaussian 03 (rev. C.02) [50] and the composite CBS-QB3 method [49]. Frequency analysis has made it possible to point out one imaginary frequency for each transition state, and IRC calculations confirmed that the transition state connected the reactant to the product.

One molecular route of decomposition of ketohydroperoxides to diones and water was found (reaction (2) in Fig. 15). The calculated rate constant is $k = 1.28 \times 10^7 T^{2.25} \exp(-44,950/RT)$ (units: cal, mol, cm^3 , s). Figure 16 displays the different channels that were found for the consumption of alkoxy radicals. These reactions are:

- a reaction of β -scission by breaking of a C–H bond, leading to a dione and an H atom (reaction (1) in Fig. 16),
- a reaction of β -scission by breaking of a C–C bond leading to an aldehyde and an alkyl radical with a carbonyl group (reaction (2) in Fig. 16),
- a reaction of β -scission by breaking of a C–C bond leading to a C_4 dione (ketone–aldehyde) and a methyl radical (reaction (3) in Fig. 16),
- a reaction of isomerization through a three-membered ring transition state (reaction (4) in Fig. 16),
- a reaction of isomerization through a five-membered ring transition state (reaction (5) in Fig. 16).

Note that reactions (4) and (5) lead to the formation of two new radicals. In this model, we considered only reactions of oxidation with O_2 for their consumption forming diones and HO_2 radicals. Alkyl radicals with a carbonyl group that are obtained from the C–C β -scission reactions (reaction type (2) in Fig. 16) are mainly consumed

by H-atom abstractions leading to the formation of ketones such as acetone, butanone, and 2-pentanone. Note that mole fractions of these three species computed with the model are slightly overpredicted (Figs. 4 and 5). This is likely due to missing routes for the consumption of ketones in the secondary mechanism.

The kinetic parameters of the five reaction channels for the consumption of alkoxy radicals are given in Table 4. Note that the kinetic parameters obtained for the reactions of β -scission (1) and (2) in Fig. 16 are in good agreement with the kinetic data obtained by Rauk et al. [62] (the model reactions are $C_2H_5-CH(-O)-CH_3 \rightarrow C_2H_5-CO-CH_3 + H$ and $C_2H_5-CH(-O)-CH_3 \rightarrow C_2H_5 + \text{acetaldehyde}$ [62]).

Another type of reaction was considered for the consumption of the $C_7H_{13}O_2$ alkoxy radical: oxidation of the radicals with molecular oxygen, forming a dione and an HO_2 radical. The kinetic parameters that were used are from Atkinson et al. [63] (the model reaction is $i-C_3H_7O + O_2 \rightarrow \text{acetone} + HO_2$ and $k = 9.04 \times 10^9 \text{ (cm}^3 \text{ mol}^{-1} \text{ s}^{-1}) \times \exp(-400 \text{ (cal mol}^{-1})/RT)$).

Note that these new reactions have almost no effect to the global reactivity. The computed mole fraction profile of $C_7H_{14}O_3$ ketohydroperoxides is compared with the signal at m/z 146 obtained using SVUV photoionization mass spectrometry in Fig. 10. There is very good agreement between the shapes of the two profiles. Note that ketohydroperoxides decompose quasi-integrally through the breaking of the O–O bond and that the direct elimination of water-forming diones does not play a significant role.

Computed and experimental mole fractions of 2,4- and 3,5-heptadiones are displayed in Fig. 10. Mole fractions are underpredicted by the model by a factor of about 40. The kinetic analysis of the model shows that the $C_7H_{13}O_2$ alkoxy radicals quasi-exclusively react by β -scission by breaking of a C–C bond and almost not by β -scission of a C–H bond or by oxidation with O_2 .

As far as diones are concerned, they mainly come from the direct elimination of water from ketohydroperoxides and from the decomposition by breaking of C–H bonds of $C_7H_{13}O_2$ alkoxy radicals. As an example, at 650 K, for the 3,5-heptadione, the direct

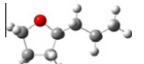
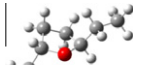
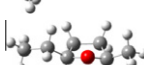
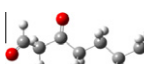
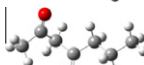
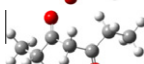
Table 1

List of species detected using gas chromatography.

Name	Global formula	Mass (amu)	Ionization energy (eV) [48]
Methane	CH ₄	16	12.61
Acetylene	C ₂ H ₂	26	11.40
Carbon monoxide	CO	28	14.01
Ethylene	C ₂ H ₄		10.51
Ethane	C ₂ H ₆	30	11.52
Propene	C ₃ H ₆	42	9.73
Carbon dioxide	CO ₂	44	13.78
Acetaldehyde	C ₂ H ₄ O		10.23
1,3-Butadiene	C ₄ H ₆	54	9.07
1-Butene	C ₄ H ₈	56	9.55
Acrolein	C ₃ H ₄ O		10.11
Propanal	C ₃ H ₆ O	58	9.96
Acetone	C ₃ H ₆ O		9.70
Acetic acid	C ₂ H ₄ O ₂	60	10.65
1,3-Pentadiene	C ₅ H ₈	68	8.59
1-Pentene	C ₅ H ₁₀	70	9.49
Methyl vinyl ketone (3-buten-2-one)	C ₄ H ₆ O		9.65
2-Butenal	C ₄ H ₆ O		9.74
Butanal	C ₄ H ₈ O	72	9.82
Butanone	C ₄ H ₈ O		9.52
Propanoic acid	C ₃ H ₆ O ₂	74	10.44
1-Hexene	C ₆ H ₁₂	84	9.44
1-Penten-3-one	C ₅ H ₈ O		9.50
2-Pentanone	C ₅ H ₁₀ O	86	9.38
Pentanal	C ₅ H ₁₀ O		9.74
1,4-Heptadiene	C ₇ H ₁₂	96	NA
1-Heptene	C ₇ H ₁₄	98	9.35
2-Heptene	C ₇ H ₁₄		8.84
1-Hexen-3-one	C ₆ H ₁₀ O		NA
Heptane	C ₇ H ₁₆	100	9.93
2-Hexanone	C ₆ H ₁₂ O		9.35
Dihydro-5-methyl-2(3H)-furanone	C ₅ H ₈ O ₂		NA
2-Ethyl-5-methylfuran	C ₇ H ₁₀ O	110	NA
2-Heptanone	C ₇ H ₁₄ O	114	9.3
3-Heptanone	C ₇ H ₁₄ O		9.1
2-Propyl-tetrahydrofuran	C ₇ H ₁₄ O		9.12 ^a
2-Ethyl-5-methyltetrahydrofuran	C ₇ H ₁₄ O		9.05–9.08 ^a
2-Methyl-4-propyloxetane	C ₇ H ₁₄ O		NA
2,4-Diethyloxetane	C ₇ H ₁₄ O		NA
2-Ethyl-3-propyloxirane	C ₇ H ₁₄ O		NA
2-Butyl-3-methyloxirane	C ₇ H ₁₄ O		NA
Tetrahydro-5-methyl-2-furanmethanol	C ₆ H ₁₂ O ₂	116	NA
2,4-Heptadione	C ₇ H ₁₂ O ₂	128	9.40 ^a
3,5-Heptadione	C ₇ H ₁₂ O ₂		9.37 ^a

^a The ionization energies of these species were not available in [48] and were calculated in this work (see Table 2) using the composite CBS-QB3 method [49] in Gaussian03 [50].

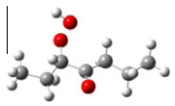
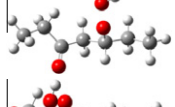
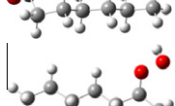
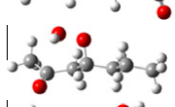
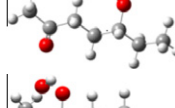
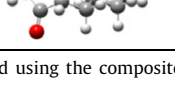
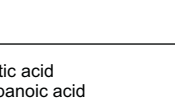
Table 2Calculated ionization energies (eV) of most expected isomers for *m/z* 114 and 128 deriving from *n*-heptane.

<i>m/z</i>	Name	Structure	Calculated ionization energy (eV)
114	<i>n</i> -Propyltetrahydrofuran		9.12
	2-Methyl-5-ethyltetrahydrofuran (trans)		9.05
	2-Methyl-5-ethyltetrahydrofuran (cis)		9.08
128	1,3-Heptadione		9.32
	2,4-Heptadione		9.40
	3,5-Heptadione		9.37

Note: Calculations were performed using the composite CBS-QB3 method [49] in Gaussian03 [50].

Table 3

Calculated ionization energies (eV) of some ketohydroperoxide isomers (m/z 146) deriving from *n*-heptane.

Name	Structure	Calculated ionization energy (eV)
4-Heptanone-3-hydroperoxide		8.87
3-Heptanone-5-hydroperoxide		9.31
1-Heptanal-3-hydroperoxide		9.32
1-Heptanal-2-hydroperoxide		9.35
2-Heptanone-4-hydroperoxide		9.57
2-Heptanone-5-hydroperoxide		9.67
2-Heptanone-3-hydroperoxide		10.65

Note: Calculations were performed using the composite CBS-QB3 method [49] in Gaussian03 [50].

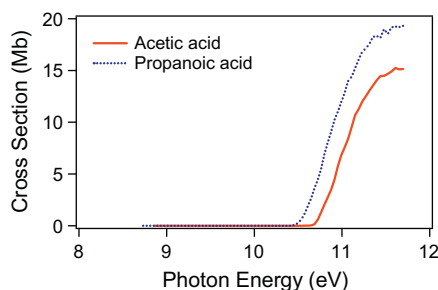


Fig. 14. Evolution of the cross sections of acetic and propanoic acids as a function of the photon energy (this work).

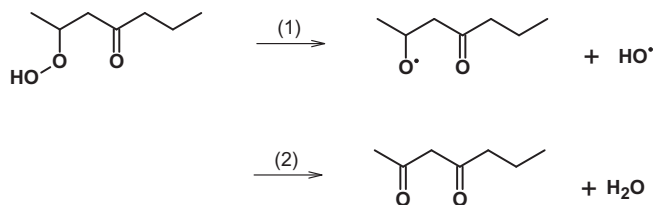


Fig. 15. Reactions that were considered for the decomposition of ketohydroperoxide species.

elimination of water represents 25% of the formation of this species, whereas the β -scission pathway is 75%.

4.3. Modeling of the formation of acetic and propanoic acids

To our knowledge, no reaction pathway has yet been proposed in the literature to explain the formation of large amounts of

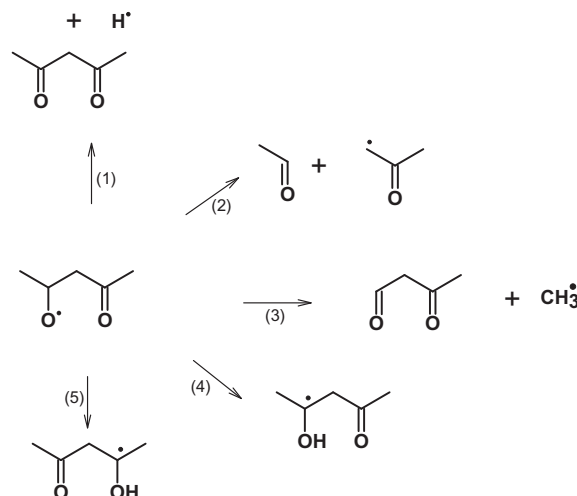


Fig. 16. Possible reactions for a model $C_5H_9O_2$ alkoxy radical. The reaction pathways were found from quantum calculations and were used to describe the fate of the $C_7H_{13}O_2$ alkoxy radicals involved in the model of the oxidation of *n*-heptane.

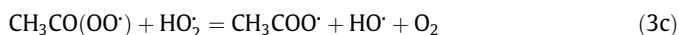
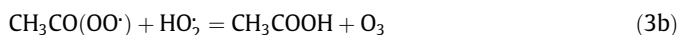
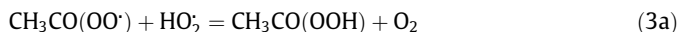
Table 4

Kinetic parameters of the reactions of consumption of a $C_5H_9O_2$ alkoxy radical.

Reaction in Fig. 16	A	n	E_a (cal mol $^{-1}$)
(1)	2.43×10^9	1.42	19,380
(2)	2.90×10^{12}	0.38	11,080
(3)	1.02×10^{12}	0.72	15,480
(4)	0.452	3.84	16,850
(5)	74.92	3.49	19,800

Notes: Kinetic parameters were obtained performing quantum calculations. Units: cal, mol, cm 3 , s.

carboxylic acids during the low-temperature oxidation of alkanes related to combustion reactions. But the formation of acids has already been investigated in atmospheric chemistry studies [64]. Besides the oxidation of olefins with ozone, an important proposed pathway is by reaction of peroxyacyl radicals with HO_2 radicals [65]. Peroxyacyl radicals are obtained from aldehydes by H-abstraction followed by an addition to oxygen molecules. The following reactions have been considered in the present model:



The kinetics of reactions (3a)–(3c) has been experimentally and theoretically investigated by Le Crâne et al. [66]: a total rate constant of 9×10^{12} cm 3 mol $^{-1}$ s $^{-1}$ has been derived from this reaction at 298 K, and it was shown that the formation of peracetic acid and oxygen was the dominant channel, followed by channel 3b with a branching ratio of 20%, and much less importantly by channel 3c. Reaction (3b) was added to the model generated using software EXGAS [31] (reactions (1), (2), and (3a) were already included in the generated model) with the rate constants proposed by Le Crâne et al. [66]. In reaction (3b), the formation of ozone was replaced by that of an oxygen molecule and an O-atom. The same type of mechanism was considered for the formation of propanoic acid

from propanal. Figure 5 displays the comparison between computed and experimental mole fractions of acetic and propanoic acids. Computed data are in relatively good agreement with the experimental data obtained using SVUV-PIMS.

4.4. Modeling of other minor species

The original EXGAS-generated *n*-heptane mechanism did not contain formation routes for some experimentally observed species: methyl vinyl ketone, 2-butenal, 1,3-pentadiene, 1,4-heptadiene, and 2-hexanone.

Methyl vinyl ketone is assumed to be a product of metathesis reactions of higher ketones, corresponding rate constants were taken from the LLNL mechanism for 2-pentanone metathesis and an analogy was made for 2-hexanone and 2-heptanone metatheses. 2-Butanal is assumed to be formed via higher ketohydroperoxide ($C \geq 5$) decomposition; again, LLNL rate constants were assigned to these reactions.

1,3-Pentadiene is assumed to be formed via hexane metathesis ($C_6H_{12} + R \cdot = RH + C_5H_8 + \cdot CH_3$) by analogy with the metathesis of C_5 olefin leading to formation of 1,3-butadiene, described in our mechanism. On the other hand, the C_7 diene (1,4-heptadiene) is assumed to be a product of the recombination of $\cdot C_3H_5$ and $\cdot C_4H_7$ radicals; given the uncertainty in the rate constant of this combination reaction, a collision limit ($1 \times 10^{14} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$) was assigned for this reaction, resulting in a slight overprediction at temperatures above 800 K.

2-Hexanone is assumed to be produced via beta scission from $C_6H_{12}OOH$, a rate constant of $1.90 \times 10^9 \exp(-7500/RT)$ (units cal, mol, cm^3 , s) was assigned by analogy with the beta scission of $C_5H_{10}OOH$ forming 2-pentanone.

In addition to these, an acetone submechanism was adopted as described in Pichon et al. [67].

4.5. Comparison of experimental and computed data

Mole fractions computed with the model were compared with experimental data (Figs. 2–9). Simulations were also performed using the model developed by LLNL [61]; the overall agreement is similar to that obtained with the EXGAS model. The comparison is available in the Supplemental data. Figure 2 displays a comparison of computed and experimental data for the reactants (*n*-heptane and oxygen). The model reproduces well the reactivity over the whole range of temperatures. The agreement is also very good for small species such as carbon oxides, methane, acetylene, ethylene, and water (Fig. 3). The formation of ethane is slightly overestimated in the high-temperature region, whereas the opposite trend is observed for propene (Fig. 3).

As far as C_1 – C_6 oxygenated species are concerned (Figs. 4 and 5), the agreement is satisfactory for acetaldehyde, butanal, and pentanal. It deteriorates slightly for methanol, formaldehyde, acrolein, propanal, and 2-pentanone. According to the kinetic analysis of our model, the overprediction of formaldehyde is not due to lumped reactions, as this species is only formed from reactions included in the C_0 – C_2 reaction base. Note that mole fractions of these species are also overpredicted by the LLNL model. For propanal and acrolein, the overprediction of the mole fractions is due to the rules of decomposition of species that were used in the model to simplify the secondary mechanism. This is also due to the simplifications in the secondary mechanism. Note that 1-penten-3-one and 1-hexen-3-one are not included in the model.

For C_4 – C_7 hydrocarbons (Figs. 6 and 7), the agreement is satisfactory for 1-butene, 1,3-butadiene, 1-pentene, 1,3-pentadiene, 1-hexene, and 1,4-heptadiene. As far as heptene isomers are concerned, only comparisons for 1- and 2-heptenes were possible, as 3-heptene was not detected in experiments. Computed mole

fractions of 1-heptene are underpredicted, whereas the opposite trend is observed for 2-heptene. Note that mole fractions computed with the LLNL model are in correct agreement for 1-heptene but overpredicted for 2-heptene.

The comparison of computed and experimental mole fractions of $C_7H_{14}O$ cyclic ethers is displayed in Fig. 8. The comparison is only displayed for cyclic ethers observed in experiments (2-propyltetrahydrofuran, 2-ethyl-5-methyltetrahydrofuran, 2-methyl-4-propyloxetane, 2,4-diethyloxetane, 2-ethyl-3-propyloxirane, 2-butyl-3-methyloxirane). The agreement is overall satisfactorily for four- and five-membered ring cyclic ethers. The model overpredicts the formation of three-membered ring cyclic ethers. Note that the formation of three- and four-membered ring cyclic ethers at high temperatures predicted by the model is not observed in the experiments (whereas it is in the case of five-membered ring cyclic ethers). This is likely due to mole fractions being below the detection limit (~ 10 – 20 ppm). A comparison is not given for dihydro-5-methyl-2(3h)-furanone and 2-furanmethanol, tetrahydro-5-methyl as routes of formation and consumption are not included in the model.

The comparison for 2- and 3-heptanone is displayed in Fig. 9 (4-heptanone was not detected in experiments). Note that heptanones were initially represented in a lumped species; these were delumped into 2-, 3- and 4-heptanones. The computed mole fractions are significantly overpredicted by the model. This is mostly due to the large uncertainty in activation energy resulting from the $C_7H_{15}OO \cdot = C_7H_{14}OOH$ isomerization rate constants. Figure 9 also shows the comparison for all $C_7H_{14}O$ (heptanal, heptanones and cyclic ethers). Computed mole fractions are underestimated by a factor of 2.

4.6. Kinetic analysis of the model

Flux analyses at 650 K and 850 K are represented in Figs. 17 and 18. *n*-Heptane mainly decomposes into 2- and 3-heptyl radicals via H-atom abstraction (71.4% at 650 K and 64.6% at 850 K). On the other hand, 1- and 4-heptyl radicals are produced to a lesser extent (28.6% at 650 K and 34.1% at 850 K). Decomposition pathways of the most abundant fuel radicals are shown in Figs. 17 and 18. As is typical in low-temperature alkane chemistry (Fig. 17), the fuel radical adds to molecular oxygen and goes through $ROO \cdot = \cdot QOOH$ isomerization, which eventually results in the formation of C_7 ketohydroperoxides and $C_7H_{14}O$ species (heptanal, heptanones, and cyclic ethers, as shown). At higher temperatures (850 K; see Fig. 18), heptyl radicals mainly decompose into smaller species through reactions of β -scission. Note that reactions of oxidation (to heptenes plus HO_2) and of addition to O_2 still play a role at this temperature.

5. Conclusions

The experimental study of the oxidation of *n*-heptane was performed in a jet-stirred reactor and reaction products were analyzed using two complementary methods: gas chromatography and SVUV photoionization mass spectrometry. A model for the low- and high-temperature oxidation of *n*-heptane was generated using the software EXGAS. The comparison of experimental and computed data is satisfactorily, keeping in mind that most discrepancies in the mole fractions of intermediates at low temperature are due to some globalization in the secondary mechanism to limit the size of the model.

Important intermediates in the low-temperature oxidation of hydrocarbons were detected in this study. The formation of ketohydroperoxides was observed using SVUV-PIMS, confirming the previous identification of this type of species in the low-temperature oxidation of *n*-butane. The computed profile matched the experimental mole fraction profile well.

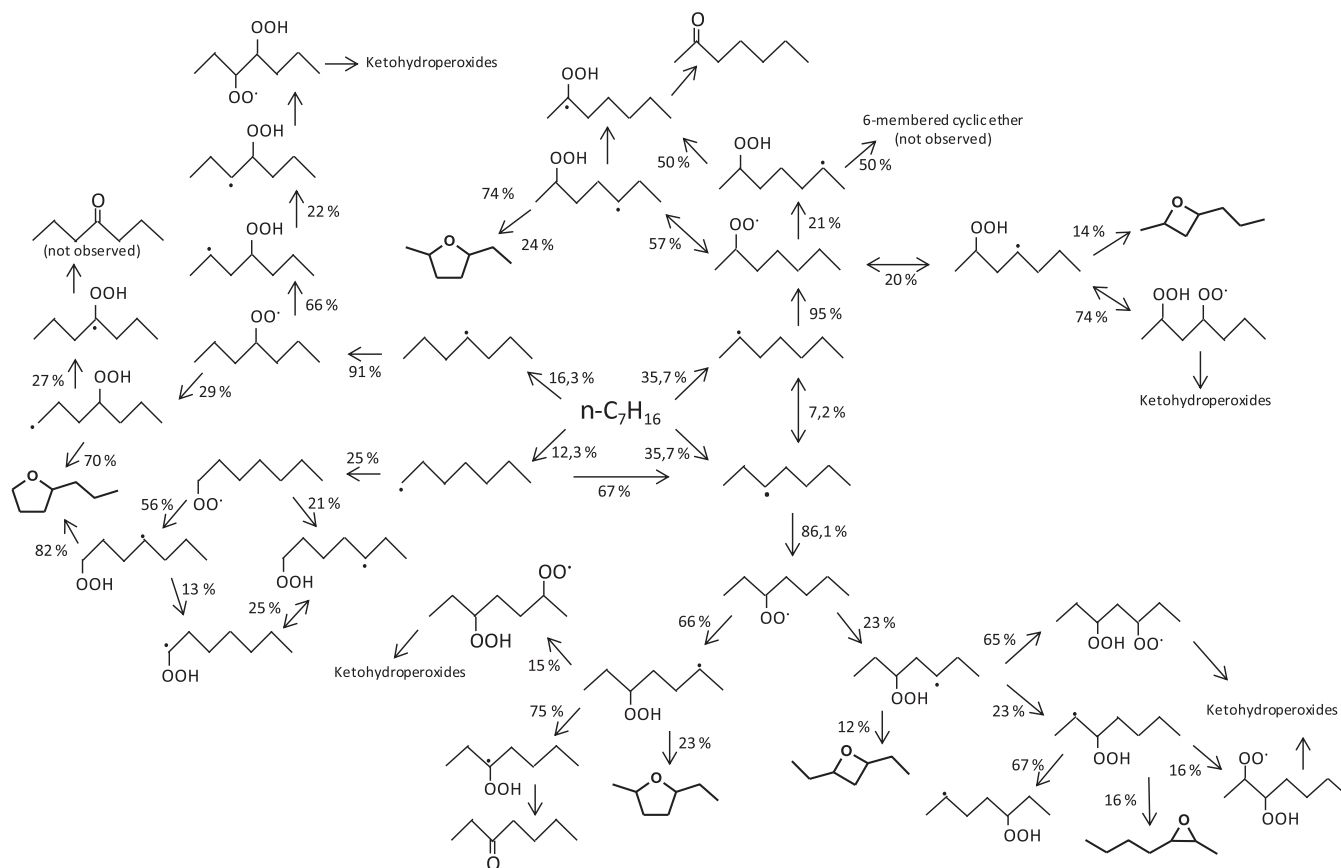


Fig. 17. Rate of production analysis performed at low temperature (650 K) using the *n*-heptane oxidation model developed in this study.

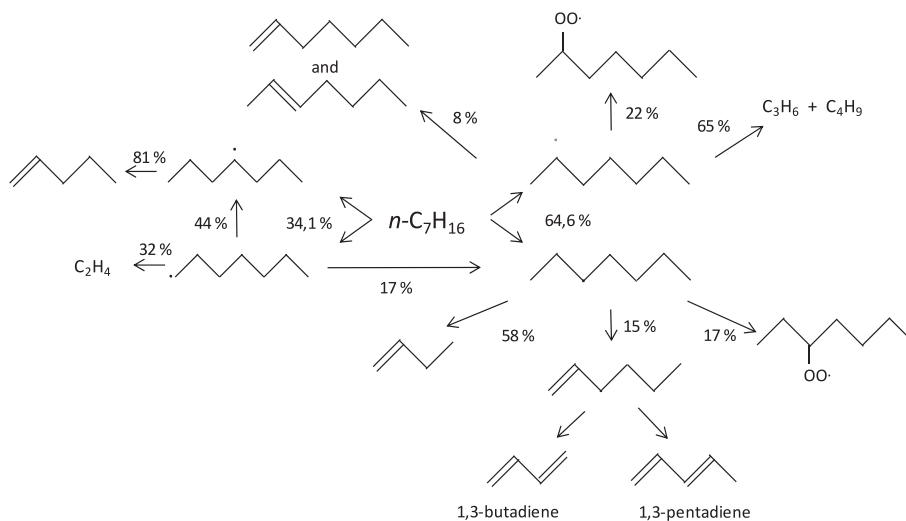


Fig. 18. Rate of production analysis performed at high temperature (850 K) using the *n*-heptane oxidation model developed in this study.

The formation of species with two carbonyl groups was also observed using both analytical techniques. Gas chromatography analyses allowed the separation of two isomers: 2,4- and 3,5-heptadiones. The gas chromatography mole fraction profiles of these two species were unusual because of the presence of a sharp peak at the lowest temperatures (as for ketohydroperoxides). This was likely due to the decomposition of ketohydroperoxides during their transfer in the line between the reactor and the gas chromatographs.

New reaction pathways were added in the model to account for the formation of these species from the ketohydroperoxides; however, the result was not successful, as the computed mole fractions are still underpredicted by a factor of 40.

The presence of organic acids, acetic and propanoic acid, was observed. The two sets of experimental data obtained with the two analytical techniques show differences. As for diones, these species are likely to be formed from the decomposition of

ketohydroperoxides in the heated transfer line between the reactor and the gas chromatograph. A route of formation of acetic acid from acetaldehyde was included in the present model.

This study shows that the chemistry of hydrocarbons at low temperatures is responsible for the formation of many species with low individual concentrations and that the scheme of the low-temperature oxidation of hydrocarbons that is included in the current models still needs to be refined to better account for the formation of minor pollutants observed in the experiments. Experimental studies of model molecules smaller than *n*-heptane, along with quantum calculations and modeling studies, are still necessary for a better understanding of the low-temperature oxidation chemistry of hydrocarbons.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2012.07.008>.

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